Meng Wang

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Research Interests

Reliable multimodal foundation models and their applications in healthcare, education, and science, including uncertainty estimation, human-AI interaction, agent systems, LLM post-training, and continual learning.

Work Experience

Research Assistant for Medical Post-Training Hong Kong July 2025 - Present

- Led development of post-training pipelines for in-house CARES 3.0&4.0 medical multimodal large language models using supervised fine-tuning, reinforcement learning, and on-policy distillation. Conducted large-scale experiments (20k+ Nvidia H800 GPU-hours) to optimize data composition and training strategies.
- Developed a holistic framework supporting 30+ public medical datasets for unified MLLM & agent evaluation. Built and evaluated an in-house medical agent system equipped with specialized medical tools (e.g., ViT classifiers) for diagnostic accuracy, interpretability, and uncertainty estimation.
- Investigated continual post-training via on-policy self-distillation, identifying the supervision regime where self-distillation improves performance. Studied uncertainty estimation and calibration for multimodal medical reasoning using logit distributions and GRPO alignment.

Independent Researcher at Interplay Lab Rochester, NY Oct 2024 - May 2025

- Developed an LLM conversational agent for collaborative reasoning during reading activities, guided by linguistic and educational theories. Adapted the agent to the cognitive level of K–12 students with techniques like RAG, ReAct prompting, and reinforcement learning.

Research Assistant at Earth Imaging Lab Rochester, NY Jun 2024 - May 2025

- Developed generative models for unstructured seismic waves with WGAN, improving fidelity by 20%.
- Designed a physics-informed neural network (iRADNet) to solve the inverse Radon transform for seismic wave migration, demonstrating 22.1% performance increase and 98.4% speed-up than traditional ISTA.

Education

2023 - 2025 M.S. in Computer Science at University of Rochester (GPA 3.91)

2019 - 2023 B.S. in Computing at The Hong Kong Polytechnic University (GPA 3.50)
II-1 Honor, *Dean's List 2019-20*, *URIS Scholarship 2021-23*
Capstone: Compressed Video Super-Resolution for Streaming Media.

Publications

[1] **M. Wang**, H. Zhao, W. Liu, L. Yang, G. Liu, H. Guo, G.-S. Xie, G. Meng, H. Liu, and F. Zhu, *Denser ≠ better: Limits of on-policy self-distillation for continual post-training*, Jul. 2026. arXiv: 2607.01763 [cs.LG]. [Online]. Available: https://arxiv.org/abs/2607.01763.

[2] R. Feng, H. Zhao, S. Tang, G. Liu, S. Lai, **M. Wang**, S. Zhuang, Y. Zha, C. Ma, and R. Yu, "Lookahead-GCG: Improving universal multi-model optimization-based jailbreaking attacks via stochastic nesterov optimization," in *Forty-third International Conference on Machine Learning*, to appear, 2026. [Online]. Available: https://openreview.net/forum?id=49ZO681M7H.

[3] S. Ma, F. Zhu, Y. Guo, **M. Wang**, W. Liu, Z. Zhong, X.-Y. Zhang, and C.-L. Liu, "Happy++: Toward stable and unified continual generalized category discovery," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 48, no. 11, pp. 15 143–15 160, 2026. DOI: 10.1109/TPAMI.2026.3721625.

[4] H. Wang, **M. Wang**, E. Farhadi, A. Shahab, Y. Li, H. Du, C. Rose, H. He, and Z. Bai, "Small arguments, big ideas: Fostering children's collaborative reasoning with LLM agents," in *Proceedings of the 2026 Annual Meeting of the International Society of the Learning Sciences*, to appear, 2026.

[5] H. Zhao†, **M. Wang**†, F. Zhu, W. Liu, B. Ni, F. Zeng, G. Meng, and Z. Zhang, "VTCBench: Can vision-language models understand long context with vision-text compression?" In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*, to appear, 2026.

[6] H. Zhao, F. Zhu, H. Guo, **M. Wang**, R. Wang, G. Meng, and Z. Zhang, *MLLM-CL: Continual learning for multimodal large language models*, Jun. 2025. arXiv: 2506.05453 [cs.CL]. [Online]. Available: https://arxiv.org/abs/2506.05453.

[7] **M. Wang** and T. Olugboji, "Deep algorithm unrolling for seismic migration," in *2025 IEEE Conference on Computational Imaging Using Synthetic Apertures (CISA)*, 2025, pp. 1–5. DOI: 10.1109/CISA64343.2025.11091556.

Skills

Programming	Python, Java, JavaScript
ML/LLM	PyTorch, HFTransformers, HFDatasets, Lightning
OSS contributions	modelscope/evalscope, huggingface/datasets, hiyouga/LlamaFactory
Training Frameworks	verl-project/verl, modelscope/ms-swift, Moenupa/LyCIL
Systems	Linux, Docker, Podman, Enroot, Slurm, distributed GPU training

Projects

Continual Post-Training via Self-Distillation Moenupa/SDPO-CL

Studied SDPO for continual post-training in comparison to GRPO. On-policy self-distillation in continual post-training accelerates in-domain specialization but fails to prevent forgetting and can collapse in out-of-distribution scenarios, indicating that on-policy data alone is insufficient for continual learning.

VTCBench: Benchmarking Vision-Text Compression Moenupa/VTCBench

Developed the first VTC benchmark evaluating VLMs on long-context understanding. Designed retrieval, associative reasoning, and multi-turn memory tasks to decouple modality factors and quantify performance trade-offs, length fragility, and spatial sensitivity under compressed visual representations.

Real-Time Image Search with Natural Language for Unsplash Moenupa/clip-image-search

Developed a CLIP-based multimodal retrieval system supporting real-time fine-grained natural language image search over the Unsplash Lite dataset, deployed on AWS SageMaker.

Human-Machine Collaborative Driving URIS-AutoVehicle/APF-Autonomous-Driving

Led a three-person team in developing an artificial-potential-field algorithm for shared-control autonomous driving. Evaluations in CARLA simulations & ROS vehicles demonstrate a 3% improvement over MPC baseline in reducing collision rate and improving trajectory smoothness.

Compressed Video Super-Resolution for Streaming Media Moenupa/mmediting

Extended BasicVSR with a pretrained image-super-resolution preprocessor to better restore sharp, synthetic artifacts (e.g., captions, edits, effects, etc.) in streaming media. Evaluated the model on streaming videos collected from YouTube, achieving a 2.7 dB increase in PSNR and a 1.6% increase in SSIM than BasicVSR.